

Joint variational autoencoders for multimodal imputation and embedding

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Single-cell multimodal datasets have measured various characteristics of individual cells, enabling a deep understanding of cellular and molecular mechanisms. However, multimodal data generation remains costly and challenging, and missing modalities happen frequently. Recently, machine learning approaches have been developed for data imputation but typically require fully matched multimodalities to learn common latent embeddings that potentially lack modality specificity. To address these issues, we developed an open-source machine learning model, Joint Variational Autoencoders for multimodal Imputation and Embedding (JAMIE). JAMIE takes single-cell multimodal data that can have partially matched samples across modalities. Variational autoencoders learn the latent embeddings of each modality. Then, embeddings from matched samples across modalities are aggregated to identify joint cross-modal latent embeddings before reconstruction. To perform cross-modal imputation, the latent embeddings of one modality can be used with the decoder of the other modality. For interpretability, Shapley values are used to prioritize input features for cross-modal imputation and known sample labels. We applied JAMIE to both simulation data and emerging single-cell multimodal data including gene expression, chromatin accessibility, and electrophysiology in human and mouse brains. JAMIE significantly outperforms existing state-of-the-art methods in general and prioritized multimodal features for imputation, providing potentially novel mechanistic insights at cellular resolution.

Understanding of molecular mechanisms at cellular resolution provides deeper insights into cellular function, development and disease progression, but remains elusive. To this end, single-cell multimodal datasets have recently emerged by using new sequencing technologies to measure various characteristics of single cells and identify cell functions (for example, cell types). For instance, Patch-seq¹ simultaneously measures gene expression, electrophysiology (ephys), and morphology (beyond omics) of single neuronal cells such as for the mouse visual cortex over 4,000 cells from several brain cell types². Using single-cell RNA sequencing (scRNA-seq) and single-cell sequencing assay for transposase accessible chromatin (scATAC-seq), Trevino et al.³ profiles single-cell gene expression and chromatin accessibility in

the human developing brain. Thus, an integration of single-cell multimodal datasets can significantly aid in our understanding of biological mechanisms contributing to cell types and diseases through the automated discovery of inter-modal relationships.

Many methods have been developed to integrate multimodal datasets to improve predictions of cell types and cellular phenotypes (see Fig. 1a for an outline of the process). For instance, our recent DeepManReg⁴ approach performs interpretable deep manifold learning that both improves phenotype prediction and is applicable to align single-cell multimodal data. Although these approaches primarily focus on prediction, the underlying idea can be extended to predict data present in separate modalities, referred to as cross-modal

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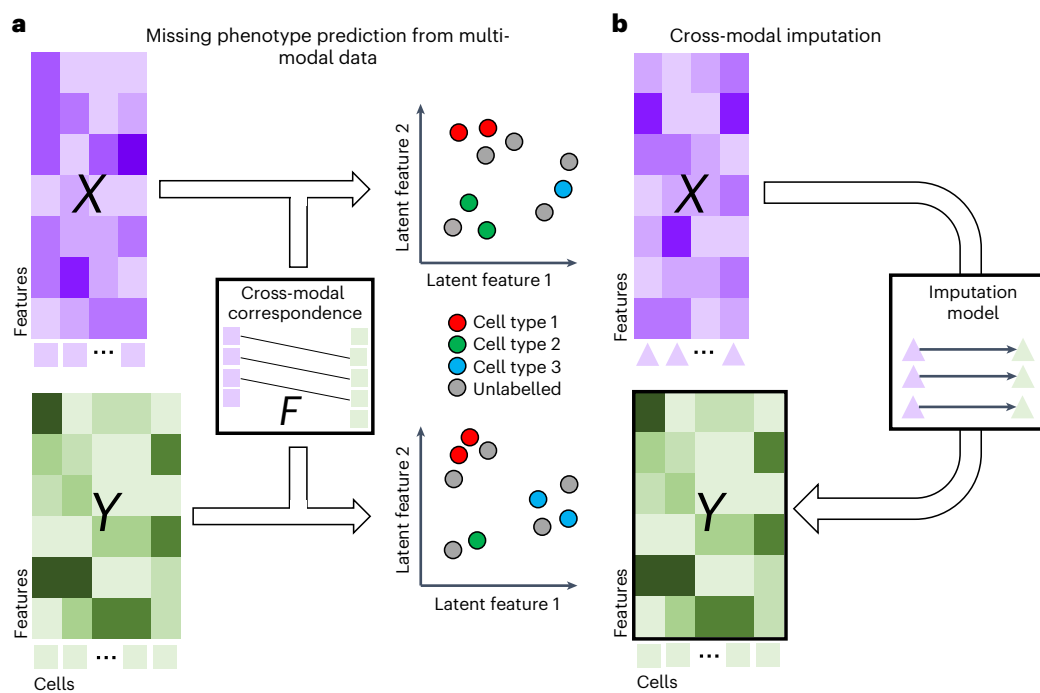


Fig. 1 | Challenges for multimodal data integration and imputation. a, Using multimodal data to effectively predict missing phenotypes (for example, cell types from multimodal single-cell data) is difficult due to heterogeneous features across modalities. Identifying similar latent spaces across modalities enables cross-modal comparison and, by extension, missing phenotype prediction. Machine learning can be used to discover similar cross-modal latent spaces

and enable comparison and phenotype prediction. **b,** Certain modalities are cost-prohibitive but may lend significant insight into biological mechanisms. As examples, scATAC-seq data for cell-type epigenomics are expensive and ephys data at single-cell resolution are difficult to produce. Imputing one modality from another by using machine learning can alleviate these constraints.

imputation below (see Fig. 1b for a visual representation). Cross-modal imputation is not new, but has been increasingly explored with the advent of deep learning. For instance, BABEL³ focuses on conversion between scRNA-seq and scATAC-seq data matrices through the use of dual autoencoders, which have the particular advantage of being reusable. Moreover, autoencoders allow for flexibility in latent space formulation. Polarbear⁶ utilizes a similar approach, focusing on a conversion network between independently generated latent spaces for each modality. Both methods, however, focus specifically on multiomics data such as scRNA-seq and scATAC-seq and thus do not consider high non-linearity from additional single-cell modalities such as ephys. Moreover, they require fully correspondent multimodal data (matched cells), limiting their capabilities with regard to data with missing modalities.

Also, analysing multimodal datasets incurs additional difficulties, including heterogeneous distributions, multi-collinearity and varying reliability. Several approaches have been thus utilized in aligning multimodal datasets (see Fig. 1b for an outline of the process). For instance, UnionCom⁷ infers correspondence information from each modality's distance matrices then uses a modified t-distributed stochastic neighbor embedding (tSNE) method for the final mapping. Maximum Mean Discrepancy Manifold Alignment (MMD-MA)⁸ minimizes an objective function to maximize distribution similarity between datasets while minimizing distortion. Single cell Graph-Linked Unified Embedding (ScGLUE)⁹ uses an autoencoder model to map different modalities onto the same latent space. Single cell Deep learning model for ATAC-Seq and RNA-Seq Trajectory integration (ScDART)¹⁰ learns a latent space and cross-modal relationships simultaneously by predicting gene activity across modalities before projecting to a single latent space. Single cell coupled Autoencoder and cycle Gene Adversarial Network (ScAEGAN)¹¹ utilizes dual autoencoders in tandem with cycle Gene Adversarial Networks (cycleGAN)¹² to provide aligned latent representations for

different modalities. Gala et al.¹³ use coupled autoencoders to perform multimodal analyses with neuronal gene expression and electrophysiological features. Certain methods rely on user-provided correspondence information to help inform alignment. For example, Cross-Linked Unified Embedding (CLUE)¹⁴ assumes completely aligned samples and introduces the impetus for aggregating correspondent latent space embeddings. ManiNetCluster¹⁵ takes a user-provided correspondence matrix as input and implements non-linear manifold alignment (NLMA) by solving an eigenvalue problem. ManiNetCluster also implements canonical correlation analysis (CCA)¹⁶ as a manifold alignment solution for multimodal data. However, if only partial correspondence information is known, existing methods are limiting, and few are designed for such a use case.

To address these issues, in this Article we describe Joint variational Autoencoders for Multimodal Imputation and Embedding (JAMIE). An outline of JAMIE's capabilities can be found in Supplementary Table 1. JAMIE trains a reusable joint variational autoencoder (VAE) model to project available multimodalities onto similar latent spaces (but still unique for each modality), allowing for enhanced inference of single-modality patterns¹⁷. To perform cross-modal imputation, data may be fed into an encoder, then the resultant latent space may be processed by the decoder of the other modality. JAMIE is able to use partial correspondence information. JAMIE combines the reusability and flexible latent space generation of autoencoders with the automated correspondence estimation of alignment methods. We compared JAMIE with state-of-the-art methods on simulation and emerging single-cell multimodal data using gene expression, chromatin accessibility, and ephys in human and mouse brains. We found that JAMIE substantially outperforms other methods (Results) and prioritized important multimodal features for multimodal imputation, also providing potentially novel mechanistic insights at cellular resolution.

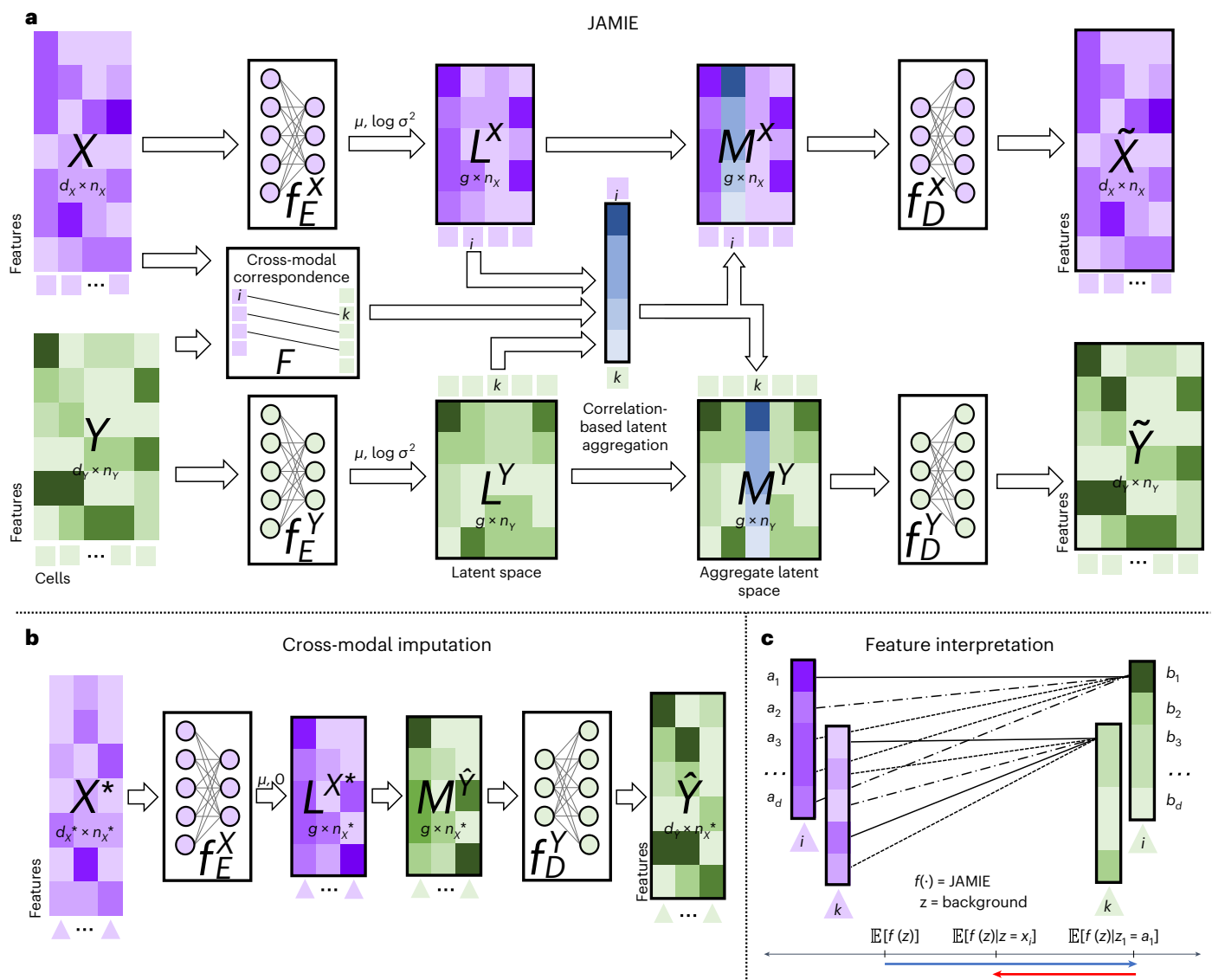


Fig. 2 | JAMIE uses VAEs with a novel latent space aggregation technique to generate similar latent spaces for each modality. a, The JAMIE model as detailed in Methods. Two data matrices X , Y are provided with an optional user-provided cross-modal correspondence matrix F . These data matrices are fed through encoding layers $f_E^X(\cdot)$, $f_E^Y(\cdot)$ which provide μ_i^X, σ_i^X and μ_j^Y, σ_j^Y for all samples i in X and j in Y , respectively. Then, $L_i^X \sim \mathcal{N}(\mu_i^X, \sigma_i^X)$ and $L_j^Y \sim \mathcal{N}(\mu_j^Y, \sigma_j^Y)$ are sampled for all i and j to produce the latent spaces L^X, L^Y . Latent spaces are aggregated using F to make M^X, M^Y . Finally, the decoding layers produce a reconstructed version of the original modalities $\tilde{X} = f_D^X(M^X)$, $\tilde{Y} = f_D^Y(M^Y)$.

The latent embeddings L^X, L^Y of the two modalities can be used in tandem for missing phenotype prediction. **b**, The trained model can be reused for cross-modal imputation through coupling of encoders and decoders from different modalities. **c**, The imputing function $f(\mathbf{x}) = f_D^Y(L^X) = \mathbf{y}$, $\mathbf{x} \in \mathbb{R}^{d_X}$, $\mathbf{y} \in \mathbb{R}^{d_Y}$ is assessed using Shapley additive explanation values¹⁸, which estimate the contribution of each input feature by selectively masking the input feature vector with the background. The symbols $a_1, a_2, \dots$ indicate the features in modality X, while $b_1, b_2, \dots$ indicate the features in modality Y. The expected value is indicated by $\mathbb{E}$.

Results

JAMIE utilizes a joint autoencoder model for data integration and imputation (Fig. 2a; see Methods for details). As input, JAMIE takes two data matrices X and Y of modalities X and Y as input. Optionally, an additional correspondence matrix F may be provided when the samples from two modalities are partially corresponding (e.g., derived from the same single cells). Encoders in JAMIE transform X and Y into latent spaces that are aggregated using available corresponding information. Decoders in JAMIE then predict reconstructions $\tilde{X}$ and $\tilde{Y}$ of the original modalities. See Methods for more details of JAMIE including training, validation and evaluation.

After training a JAMIE model, its encoder for modality X and decoder for modality Y can be used sequentially to impute from one

modality to another (Fig. 2b). Also, the latent spaces from the JAMIE model can be used for phenotype prediction. Furthermore, the use of Shapley additive explanation values¹⁸ and similar importance evaluation methods then allows us to prioritize multimodal features for imputation (Fig. 2c). These applications are further documented in 'Applications'.

Simulated multimodal data

We first tested JAMIE on simulated single-cell multimodal data⁸. The simulation data were generated by sampling from a Gaussian distribution on a branching manifold (Fig. 3a). We found that the latent embeddings of two modalities in JAMIE preserve the branching structure of the manifold while aligning the cells of the same type in either

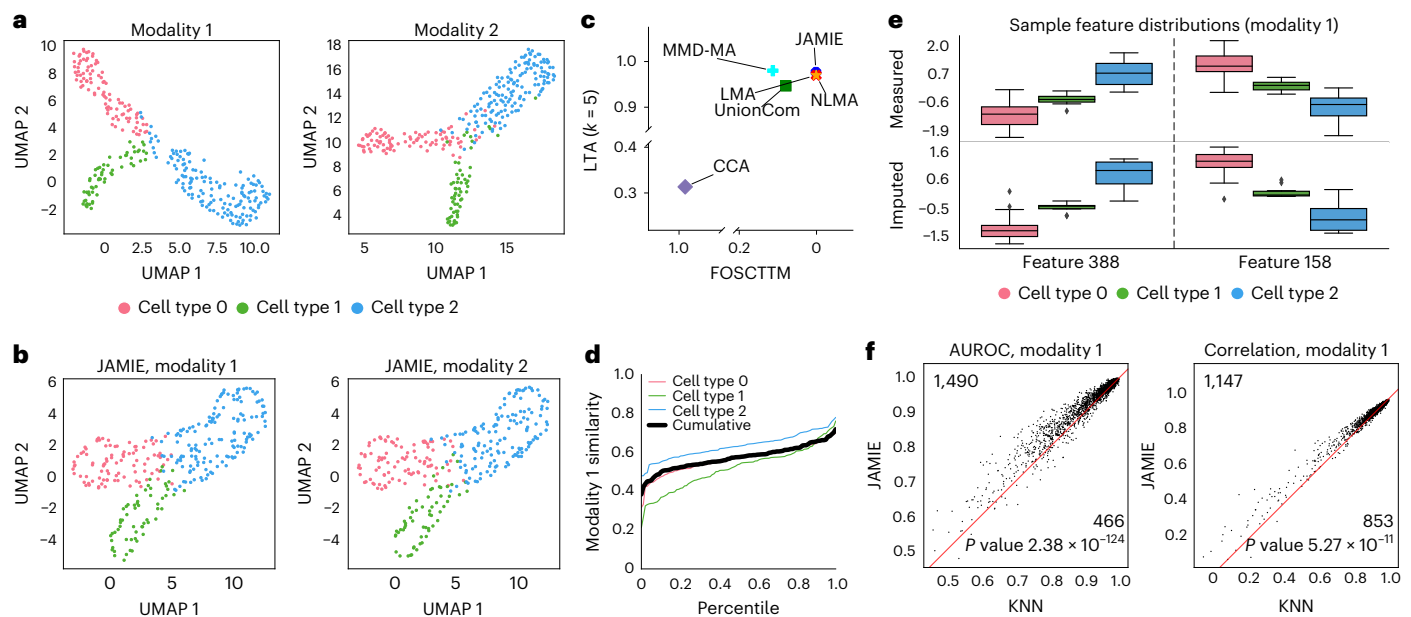


Fig. 3 | Simulated multimodal data⁸. **a**, UMAP of original spaces coloured by cell type. **b**, UMAP of JAMIE latent spaces. **c**, The fraction of samples closer to the true mean (x axis) and LTA (y axis) of JAMIE and state-of-the-art methods for cell-type separation using all correspondence information available: CCA^{15,16}, LMA¹⁵, MMD-MA⁸, NLMA¹⁵ and UnionCom⁷. **d**, Cumulative distributions of similarity (1 - JS distance) between measured and imputed feature values in modality 1. The black line indicates the average similarity across cell types, while the coloured

lines each correspond to a single cell type. **e**, Measured (top) and imputed (bottom) values of two select features in modality 1 across cell types ($n = 300$). Boxes span from the upper to lower quartiles, and a line indicates the median. Whiskers extend to extremes up to 1.5 times the interquartile range from the upper and lower quartiles. Any outliers beyond this range are plotted individually. **f**, Imputation performance for modality 1 of JAMIE versus a baseline KNN by AUROC and correlation. We utilize a two-tailed binomial test to generate P values.

modality, while also maintaining cell-type separation (Fig. 3b). To quantify the integration quality, we utilize two metrics: label transfer accuracy (LTA)^{7,19}, which measures cross-modal phenotype separation, and fraction of samples closer to the true match (FOSCTTM), which measures cross-modal alignment. More details can be found in ‘Performance evaluation’. For separating cell types, JAMIE (LTA 0.976, FOSCTTM 0.001) outperforms state-of-the-art alignment methods NLMA (LTA 0.970, FOSCTTM 0.001) in LTA and UnionCom (LTA 0.947, FOSCTTM 0.079) in both LTA and FOSCTTM (Fig. 3c).

Also, we found that the feature values imputed by JAMIE are consistent with the measurements. For instance, as shown in Fig. 2d, the imputed values of modality 1 features have high distribution similarity with the measured values across cell types with average Jensen–Shannon (JS) distance of 0.428 ± 0.097 (Fig. 3d). Two features with high similarity are highlighted with average JS distance of 0.278 and 0.281 (Fig. 3e), also showing a preservation of expression changes across cell types (that is, cell type 1 has a lower value than both other cell types). The imputation performance is compared with a baseline method through correlation and AUROC in Figs. 3f, 4f and 5f. Each dot represents a cell, and the axes are the performance for each method. The red line is the space of equal performance between methods. For imputing the first modality, we also see that JAMIE outperforms the baseline k-nearest neighbor imputation (KNN) using a two-tailed binomial test for 1,490 versus 466 features ($P < 1 \times 10^{-100}$) in terms of area under the ROC curve (AUROC) and 1,147 versus 853 features ($P < 6 \times 10^{-11}$) in terms of correlation (Fig. 3f). JAMIE exhibits similar outperformance for AUROC when imputing the second modality (Supplementary Fig. 1). We see that JAMIE is able to reliably predict features (see select cells in Supplementary Fig. 2).

We tested JAMIE on non-Gaussian simulation data²⁰ as well and found that JAMIE performed better than the majority of state-of-the-art methods (Supplementary Fig. 3). JAMIE achieved an LTA of 0.952 and FOSCTTM of < 0.001 . The second-best performing method (NLMA) had

an LTA of 0.886 and FOSCTTM of 0.002, similar to the performance of JAMIE with only 75% correspondence information provided (LTA 0.876, FOSCTTM 0.016; Supplementary Fig. 4). Further details on this dataset can be found in Supplementary Sect. 2.

Mouse neuronal gene expression and ephys

We applied JAMIE to infer cross-modal embeddings and impute gene expression and electrophysiological (ephy) data (for inhibitory neuronal types Lamp5, Serpinf1, Sst, Vip, Pvalb and Snca) neuronal cells in the mouse visual cortex (Gouwens et al.² and Fig. 4a).

We found that both gene and ephys embeddings identified by JAMIE can effectively separate those cell types (Fig. 4b). Further, JAMIE (LTA 0.944, FOSCTTM 0.002) outperforms other alignment methods in both LTA and FOSCTTM, for example, linear manifold alignment (LMA) (LTA 0.907, FOSCTTM 0.072) and UnionCom (LTA 0.887, FOSCTTM 0.124) (Fig. 4c).

Also, JAMIE performs imputation consistently across cell types and generally maintains ephys changes across cell types with an average JS distance of 0.537 ± 0.115 (Fig. 4d). Two ephys features with high similarity demonstrate this preservation with average JS distance of 0.314 and 0.316 (Fig. 4e). Finally, ephys features are imputed significantly better than the baseline KNN, with 33 of 39 features ($P < 2 \times 10^{-5}$) performing better on JAMIE for both AUROC and correlation (Fig. 4f). The imputation performance is visualized for select cells in Supplementary Fig. 2.

JAMIE was able to generalize to phenotypes beyond cell type, achieving a relatively high LTA (0.650) when predicting cortical layers compared with other state-of-the-art methods (Supplementary Fig. 5). For reference, NLMA achieved an LTA of 0.663 while LMA, the next best, achieved an LTA of only 0.515.

In addition to the mouse visual cortex, we also tested JAMIE on gene expression and ephys features in the mouse motor cortex. JAMIE maintains cell-type separation in the latent space (Supplementary Fig. 6b). JAMIE (LTA 0.899, FOSCTTM 0.002) outperforms several

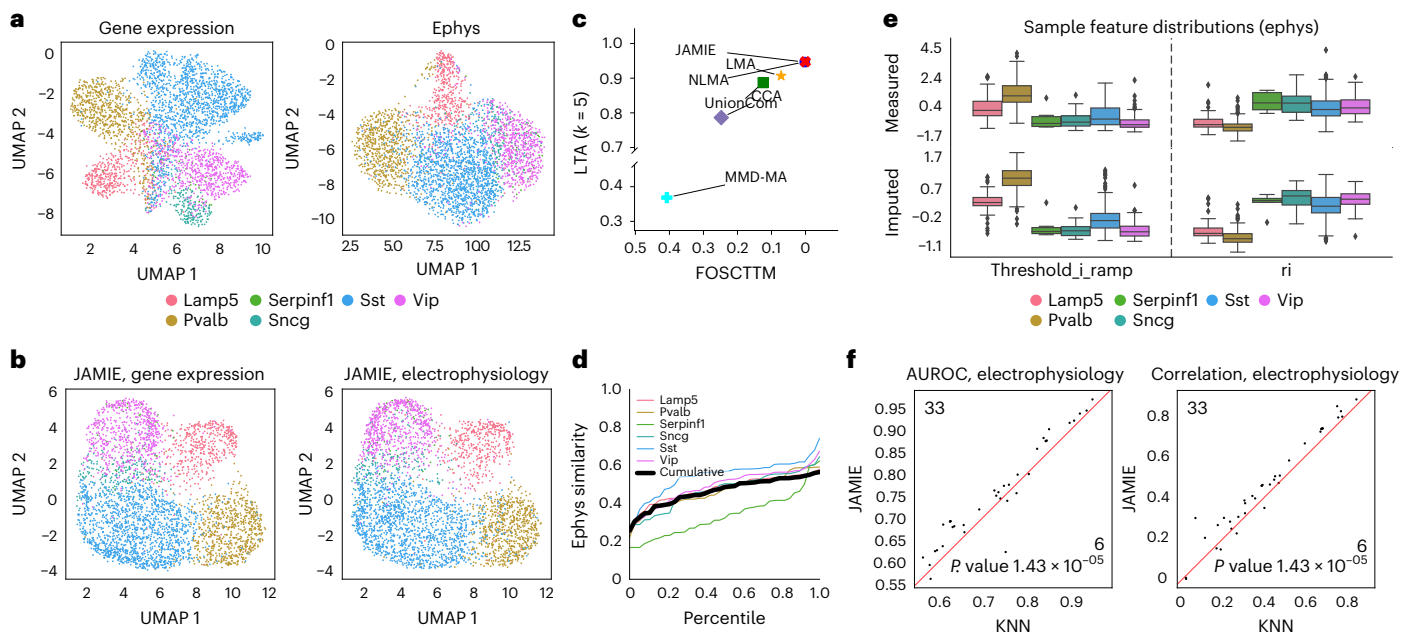


Fig. 4 | Gene expression and electrophysiological features in the mouse visual cortex². **a**, UMAP of single cells by gene expression data (left) and electrophysiological features (right), coloured by cell type (inhibitory neuronal type). **b**, JAMIE latent spaces of **a**. **c**, The fraction of samples closer to the true mean (x axis) and the LTA (y axis) of JAMIE and state-of-the-art methods for cell type separation using all correspondence information available: CCA^{15,16}, LMA¹⁵, MMD-MA⁸, NLMA¹⁵ and UnionCom⁷. **d**, Cumulative distributions of similarity (1 – JS distance) between measured and imputed electrophysiological features by gene expression. The black line indicates the average similarity across cell

types, while the coloured lines each correspond to a single cell type. **e**, Measured (top) and imputed (bottom) values of two electrophysiological features across cell types ($n = 3,654$). Boxes span from the upper to lower quartiles, and a line indicates the median. Whiskers extend to extremes up to 1.5 times the interquartile range from the upper and lower quartiles. Any outliers beyond this range are plotted individually. **f**, Performance for imputing electrophysiological features of JAMIE versus a baseline KNN by AUROC and correlation. We utilize a two-tailed binomial test to generate P values.

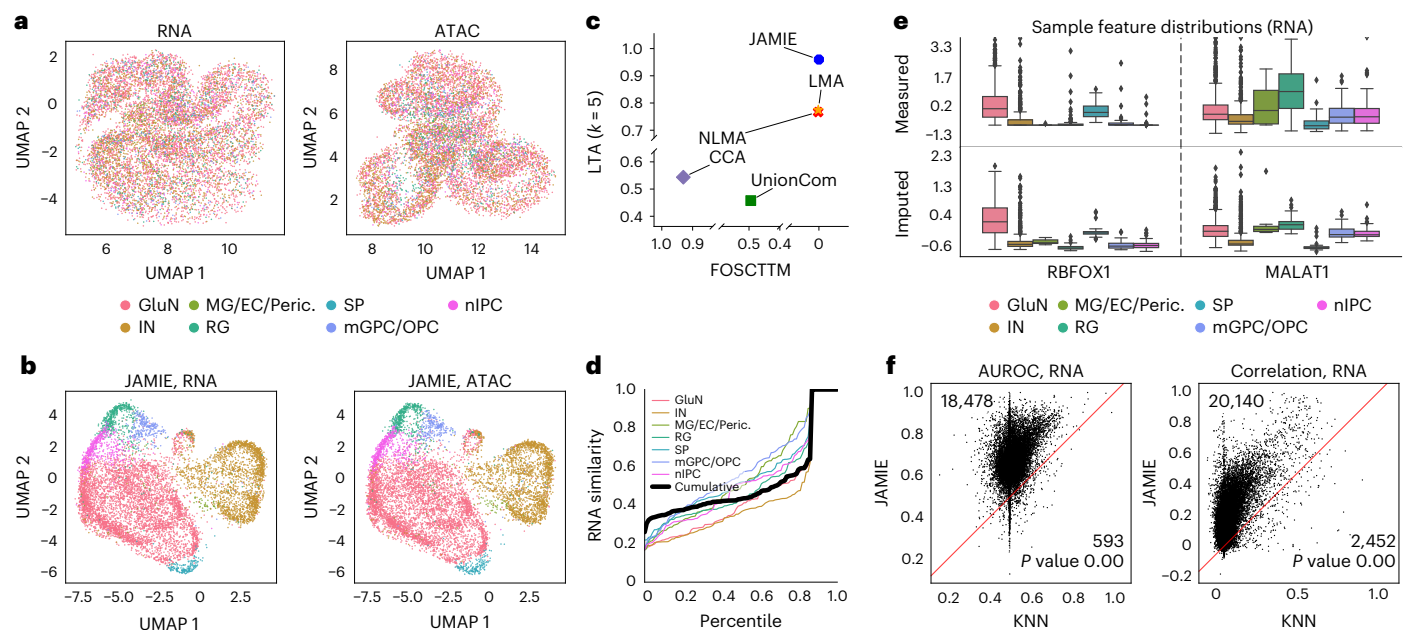


Fig. 5 | Gene expression and chromatin accessibility of single cells in the developing brain at 21 postconceptional weeks³. **a**, UMAP of single cells in the developing human brain by gene expression (left) and the accessibility activities of OCRs (right), coloured by cell type. **b**, JAMIE latent space of **a**. **c**, The fraction of samples closer to the true mean (x axis) and LTA (y axis) of JAMIE and state-of-the-art methods for cell-type separation using all correspondence information available: CCA^{15,16}, LMA¹⁵, NLMA¹⁵ and UnionCom⁷. **d**, Cumulative distributions of similarity (1 – JS distance) between measured and imputed gene expression

by chromatin accessibility. The black line indicates the average similarity across cell types, while the coloured lines each correspond to a single cell type. **e**, Measured (top) and imputed (bottom) values of two genes across cell types ($n = 8,981$). Boxes span from the upper to lower quartiles, and a line indicates the median. Whiskers extend to extremes up to 1.5 times the interquartile range from the upper and lower quartiles. Any outliers beyond this range are plotted individually. **f**, Performance for imputing genes of JAMIE versus a baseline KNN by AUROC and correlation. We utilize a two-tailed binomial test to generate P values.

methods in both LTA and FOSCTTM, for example, LMA (LTA 0.897, FOSCTTM 0.044) (Supplementary Fig. 6c). UnionCom (LTA 0.248, FOSCTTM 0.445), CCA (LTA 0.273, FOSCTTM 0.366) and MMD-MA (LTA 0.246, FOSCTTM 0.478) are unable to align the datasets.

JAMIE also preserves ephys changes across cell types, with average JS distance of 0.497 ± 0.244 (Supplementary Fig. 6d). Two ephys features with high similarity demonstrate the preserving nature of JAMIE with JS distances of 0.380 and 0.405 (Supplementary Fig. 6e). Finally, JAMIE significantly outperforms the baseline KNN using a two-tailed binomial test with 22 of 29 features for correlation ($P < 9 \times 10^{-3}$) (Supplementary Fig. 6f).

Gene expression and chromatin accessibility in human brain

To further investigate the emerging single-cell multiomics data on gene regulation, we apply JAMIE to the gene expression and chromatin accessibility data of the developing human cerebral cortex (scRNA-seq and scATAC-seq by 10x Multiome)³ (Fig. 5a). The chromatin accessibility data measure the accessibility of open chromatin regions (OCRs) by peak signals. OCRs play key epigenomic roles in regulating gene expression.

JAMIE embeddings separate cell types, in contrast to Uniform Manifold Approximation and Projection (UMAP) alone (Fig. 5a,b). In comparison of cell-type separations, JAMIE (LTA 0.959, FOSCTTM <0.001) also outperforms all other methods in both LTA and FOSCTTM compared with NLMA (LTA 0.767, FOSCTTM 0.002) and LMA (LTA 0.775, FOSCTTM 0.002) (Fig. 5c). CCA (LTA 0.544, FOSCTTM 0.930) and UnionCom (LTA 0.458, FOSCTTM 0.494) failed to align owing to the complexity of the data. We note that JAMIE provided with 75% correspondence information (LTA 0.951, FOSCTTM 0.047) and 50% correspondence information (LTA 0.936, FOSCTTM 0.106) also outperforms all other methods in terms of LTA (Supplementary Fig. 4).

Moreover, we used JAMIE to impute gene expression from OCRs (peaks) and vice versa. Imputed gene expression values are preserved across cell type with average JS distance of 0.484 and s.d. of 0.245 (Fig. 5d). We observe wider distributions in the measured data when compared with the imputed data, probably leading to relatively high JS distance with a large number of cell types. We highlight two genes with specifically high similarity, having JS distance of 0.260 and 0.332 (Fig. 5e). Finally, JAMIE significantly outperforms the imputation baseline using a two-tailed binomial test, better imputing 18,478 of 19,071 genes for AUROC ($P < 1 \times 10^{-100}$) and 20,140 of 22,592 genes for correlation ($P < 1 \times 10^{-100}$) from OCRs (Fig. 5f). JAMIE exhibits similar imputation performance when imputing ATAC features from RNA (Supplementary Fig. 7). Imputation of OCRs from gene expression also outperforms the KNN baseline using the same statistical measure ($P < 1 \times 10^{-100}$) (Supplementary Fig. 7).

A strong integration method we mentioned in the introduction is scGLUE⁹, which utilizes a combined autoencoder and graph model. To facilitate a comparison between scGLUE and JAMIE, we also ran both methods on Chen et al.²¹, which was utilized in Cao et al.⁹. We found that JAMIE (LTA 0.859, FOSCTTM <0.001) outperformed scGLUE (LTA 0.854, FOSCTTM 0.038) in both LTA and FOSCTTM (Supplementary Fig. 8). NLMA (LTA 0.259, FOSCTTM 0.035) was unable to achieve cell-type separation.

Gene expression and chromatin accessibility in colon cancer

To further test imputation performance, we compared JAMIE with BABEL⁵ using scRNA-seq and scATAC-seq data in colon adenocarcinoma COLO-320DM cells (Supplementary Fig. 9)²². JAMIE significantly outperforms BABEL in imputing both modalities by a two-tailed binomial test. In particular, JAMIE was better on 12,309 ($P < 1 \times 10^{-100}$) and 13,334 ($P < 1 \times 10^{-100}$) genes by AUROC and correlation, respectively, for gene expression imputation from chromatin accessibility (scATAC-seq, OCR peaks) (Supplementary Fig. 9a). For imputing OCRs from gene expression, JAMIE was better on 28,120 ($P < 1 \times 10^{-100}$) and

49,936 ($P < 1 \times 10^{-100}$) features by AUROC and correlation, respectively (Supplementary Fig. 9b).

Biological interpretability for cross-modal imputation

To avoid the black-box nature of many deep learning models, we applied Shapley Additive Explanations (SHAP)¹⁸ to prioritize features for cross-modal imputation (see 'Latent and imputed feature interpretation'). Specifically, this analysis gives top features in one modality for imputing a given feature in another modality.

As shown in Fig. 6a, JAMIE prioritized OCRs for imputing the gene *DENND1B*, which is a gastric cancer related gene on chromosome 1²³. Further, keeping OCRs closer to the location of *DENND1B* generally results in better imputation performance. For instance, removing OCRs within 10 kb produces lower correlation than elsewhere within the chromosome, suggesting the possible capability of JAMIE to reveal the importance of genomic proximity from chromatin accessibility to gene expression.

For the developing brain, JAMIE identifies several important genes, including *MIA* and *BBC3* (both with LTA of 0.889) for contributing JAMIE embeddings to separate the cell types. *MIA* has been linked to increased risk for neurodevelopmental disorders²⁴, and *BBC3* has been linked to cell death in the adult brain²⁵. Additionally, JAMIE identifies the gene *SST* (LTA 0.423) as an important gene in the mouse visual cortex. *SST* is known to be directly related to visual discrimination²⁶. Many cell-type marker genes were found within the top 200 prioritized genes, with all possible found within the top 400.

Discussion

JAMIE is a novel deep neural network model for cross-modal estimation. It works for complex, mixed or partial correspondence multimodal data facilitated by a novel latent embedding aggregation methodology reliant on a joint VAE structure. In addition to its outperformances as above, JAMIE is also computationally efficient and requires little memory usage (Supplementary Table 2). Moreover, the pretrained model along with learnt cross-modal latent embeddings may be reused for downstream analyses.

We evaluated JAMIE using the evaluation metrics that were designed for the alignment methods such as FOSCTTM and LTA. Despite not being explicitly designed to provide common latent spaces in each modality like alignment, JAMIE's FOSCTTM remains comparable to, and in many applications better than, existing methods in this paper. In addition, LTA appears to be better under complex use cases featuring unfiltered and noisy data. This suggests that the use of modality-specific and aggregated latent spaces in JAMIE allows for flexibility in forming the latent embeddings for better separate cell types from noisy samples.

In cases with low user-provided correspondence, we found that the alignment loss becomes more important. As seen in Supplementary Fig. 10a, imputation performance is aided through the inclusion of alignment loss with low (50%) user-provided correspondence. In highly correspondent applications, the alignment loss may be disregarded to speed up computation by eliminating the computation time of $\tilde{F}$ (Supplementary Table 2).

We have demonstrated the capability of JAMIE to impute between two modalities bidirectionally. In practice, imputing a biological outcome from the source modality is likely to be more useful than imputing a source modality from a biological outcome. Examples include imputing ephys features from gene expression and gene expression from epigenomic regulation (for example, chromatin accessibility). However, our analyses for partial correspondence provide another imputation use case. As stated in the introduction, missing modalities occur frequently when measuring single-cell multimodal data owing to its process complexity. This generates data with partial correspondence, that is, with unmatched cells between modalities. In this case, we showed that JAMIE with partial correspondence is able to

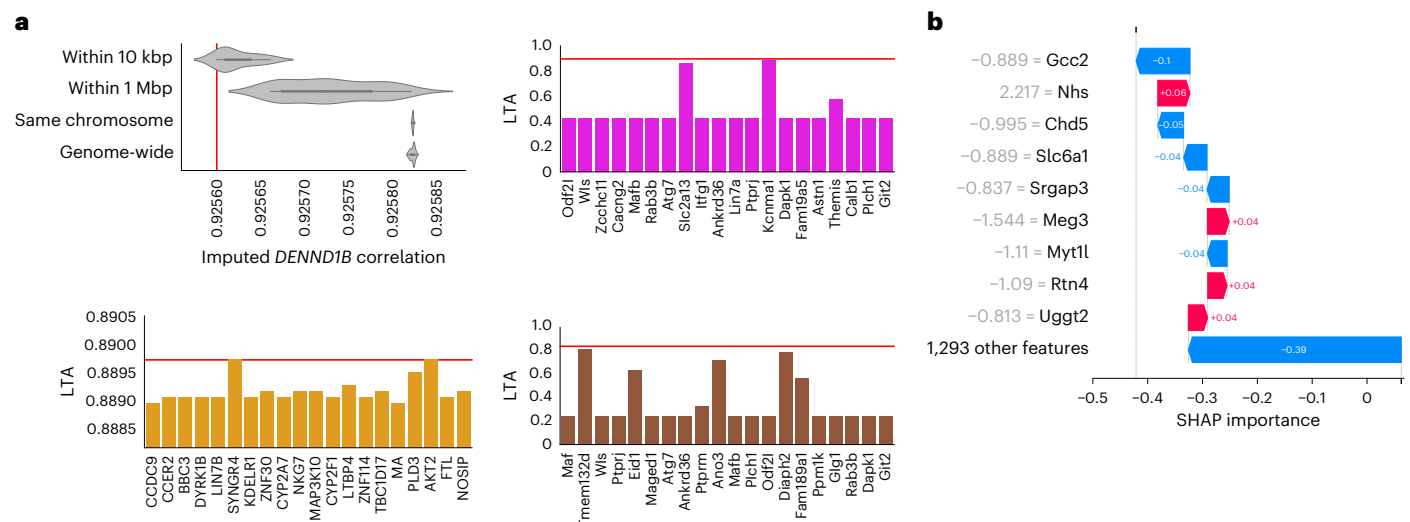


Fig. 6 | Feature prioritization for cross-modal imputation and embedding.

a, Top left: JAMIE imputation of the *DENND1B* gene with select OCRs removed in colon cancer data²². The x axis is the correlations of the imputed gene expression data with and without select OCRs removed. Aside from the same chromosome and genome-wide results, all distributions have significantly different means using a one-tailed *t* test (all *P* values $< 2 \times 10^{-10}$). Top right: LTA for JAMIE with select genes removed in the mouse visual cortex². Bottom left: LTA for JAMIE with select genes removed in the developing brain³. Bottom right: LTA for JAMIE with select

genes removed in the mouse motor cortex³¹. For each, 14 of the most significant values are displayed along with 6 randomly sampled genes for background. The red line shows the baseline value with no chromatin regions or genes removed. **b**, Waterfall plot of select important genes for imputing the ephys feature `fast_trough_t_long_square` in the mouse visual cortex Patch-seq dataset² for SST cell #601810307. Values on the y axis indicate normalized gene expression. Numbers in the body of the plot indicate SHAP values per feature. Red and blue indicate positive and negative SHAP values, respectively.

outperform a baseline KNN imputation method with full correspondence (using all samples). For instance, as shown in Supplementary Fig. 10b, JAMIE imputation using 50% of user-provided correspondence significantly outperforms the baseline by a two-tailed binomial test using all user-provided correspondence while imputing ephys features from gene expression in the mouse visual cortex ($P < 2 \times 10^{-3}$).

JAMIE started as a vanilla deterministic autoencoder but was later changed to be variational because of the superior performance of VAEs regarding interpolation in the latent space²⁷. Intuitively, the continuous latent space allows for easy sampling and interpolation²⁸. Additionally, the aggregation method proposed in ‘Aggregation of the latent spaces’ relies on the interpolation of latent representations and is improved after switching to a continuous latent space. Though VAEs perform random sampling in the latent space, the process does not add any noticeable randomness to the prediction. Given the same input, the encoder indeed can generate different latent variables owing to random sampling during training, but our decoder still reproduces very similar reconstructions. This is because the decoder is trained to map latent samples referring to the same input to very close reconstructions²⁸. On the topic of random sampling, the dropout layers do not add randomness to the prediction. They only serve to generalize the model and prevent overfitting by adding redundancy. We chose a default value of $P = 0.6$ to maximize generalizability without adversely affecting performance with default hyperparameters. From Supplementary Fig. 11, we see that dropout values of 0.4 and 0.6 have roughly equivalent mean performance while 0.8 has slightly degraded results.

We can split JAMIE into preparation and training phases. The former involves preprocessing, principal component analysis (PCA) and calculating the inferred correspondence matrix, while the latter is the joint VAE model detailed in Fig. 2. Supplementary Table 2 shows the consumed time and memory of each phase in seven datasets. For larger datasets, most computation time is spent on the preparation phase. We see that the training time scales primarily with cell count, while the peak memory usage generally scales with both cell count and number of features.

Over the course of runtime, we also found that the Kullback–Leibler (KL) divergence dictated the pace of training, as the other losses converge quickly alone (Supplementary Fig. 12). Visually, we also see that, as the KL-annealing parameter κ (see ‘Continuity of the aggregate latent spaces’) progresses, an effect can be seen in the alignment and combination losses while the reconstruction loss steadily declines, indicating that the reconstruction and KL losses dominate the training. Without KL annealing, we found that other losses were unable to converge, even with the KL loss having a relatively low weight. This was computed on our non-Gaussian simulation data²⁰.

Training VAEs is time consuming for larger datasets. Thus, prior feature selection such as the automated PCA in JAMIE helps alleviate time requirements. Data preprocessing is also crucial to avoid large or repeated features that can disproportionately shape the features of the low-dimensional embedding, owing to the use of reconstruction loss. For cross-modal imputation specifically, the diversity of the training dataset has to be carefully considered to avoid biasing the final model and negatively affecting the generalizability. JAMIE could also potentially be extended to align datasets from different sources rather than different modalities, such as gene expression measured under different conditions. Because the novel contributions of JAMIE are mainly in the processing of the latent space, many changes in structure and data type are straightforward to achieve.

Methods

Figure 2a shows the training process of JAMIE. Two numeric data matrices of modalities *X* and *Y* are used as input. $X \in \mathbb{R}^{d_x \times n_x}$ is a data matrix of modality *X* with d_x features and n_x samples. The *i*th column of *X*, that is $\mathbf{x}_i \in \mathbb{R}^{d_x}$, is the feature vector of the *i*th sample of modality *X*. $Y \in \mathbb{R}^{d_y \times n_y}$ is a data matrix of modality *Y* with d_y features and n_y samples. Similarly, the *j*th column of *Y*, that is $\mathbf{y}_j \in \mathbb{R}^{d_y}$, is the feature vector of the *j*th sample of modality *Y*. Each row of *X* and *Y* is standardized, for example, with mean of 0 and s.d. of 1. If available, a user-provided correspondence matrix $F \in \mathbb{R}_{\geq 0}^{n_x \times n_y}$ is used to aid construction of the similar latent spaces during training, where $F_{ij} = 1$ implies

one-to-one correspondence (that is, same cells different modalities) between cell i in modality X and cell j in modality Y, $F_{ij} = 0$ implies no known correspondences and $0 < F_{ij} < 1$ implies partial confidence of correspondence.

Jamie utilizes joint VAEs to learn similar latent spaces for each of the two modalities, $L^X \in \mathbb{R}^{g \times n_x}$ and $L^Y \in \mathbb{R}^{g \times n_y}$, where g is the selected dimensionality of the latent space. Higher g will lead to better reconstruction loss, but the features will not be as dense information-wise. Lower g will condense information more efficiently, but at the cost of reconstruction and imputation performance. We experimentally found $g = 32$ to be robust across multiple datasets, and use this as the default for JAMIE. During training, we utilize correlation-based latent aggregation. To learn these latent spaces, JAMIE minimizes the following losses:

$$\ell_{\text{total}} = \ell_{\text{kl}} + \alpha \ell_{\text{combination}} + \beta \ell_{\text{reconstruction}} + \gamma \ell_{\text{alignment}}. \quad (1)$$

ℓ_{kl} takes the KL divergence between the inferred distribution of our VAEs and the multivariate standard normal, aiding with the continuity of the latent space. $\ell_{\text{combination}}$ enforces similarity of correspondent samples and takes the weight α . $\ell_{\text{reconstruction}}$ is the summed mean squared error between the reconstructions and original data matrices and is assigned the weight β . $\ell_{\text{alignment}}$ uses inferred cross-modal correspondence to shape the resultant latent space and is assigned the weight γ . The weights are user defined, and their default values are detailed in 'JAMIE algorithm'. An overview of the JAMIE algorithm can be seen in Fig. 2.

The default hyperparameters work for a wide variety of cases, and are used for our applications unless otherwise mentioned, but may be customized to the user's preference.

ℓ_{kl} affects the continuity of the latent space and aids in interpolation and stability. This is particularly important for imputation applications. Although this effect is not obvious from the integrated space itself (Supplementary Fig. 13a), the imputation performance is substantially improved with this loss (Supplementary Fig. 13b).

$\ell_{\text{combination}}$ controls the focus on similarity between correspondent features across modalities. A higher weight will result in closer cross-modal cell representations at the risk of losing per-modality detail. $\ell_{\text{reconstruction}}$ controls the ability of the latent space to encode raw information from each modality. A larger weight will make cell representations more adherent to their single-modal representations at the cost of cross-modal representation similarity of correspondent samples. The effects of these losses are very pronounced when removed from the training process, and result in lower integrated space reliability (Supplementary Fig. 13a).

Finally, $\ell_{\text{alignment}}$ is primarily helpful for low-correspondence applications where the shape of the latent space is not clear from context. A higher weight will increase the similarity of cross-modal correspondent cell representations by changing the shape of the resultant latent space. This can come at the cost of cell-type mixing and cluster accuracy but can result in better imputation performance in low-correspondence applications (Supplementary Fig. 10a).

The latent dimension g can be lowered for more compressed representations, potentially at a loss in accuracy (Supplementary Fig. 14a). This can be useful in applications that present overfitting, normally when the dataset has very few features or samples. The number of principal components (PCs) used in the JAMIE model can be adjusted to change the model size, which can speed up calculation at the cost of performance (Supplementary Fig. 14b). This may also be lowered to avoid overfitting.

Distribution of latent space features

Intermediate feed-forward neural networks are utilized to generate distributions for each feature in the g -dimensional latent space for each modality. These are referred to below as encoders, denoted

$f_E^X(\cdot), f_E^Y(\cdot)$ for each modality. The variational encoders have four hidden layers. The encoder of modality X transforms the feature dimension as follows: $n_x \rightarrow 2n_x \rightarrow n_x \rightarrow g \rightarrow 2g$. For modality X, the top half of the fourth layer output corresponds to means and the bottom half corresponds to log variance per sample and feature as in equation (2a). The encoder is similar for Y. Batch normalization and leaky Rectified Linear Unit (ReLU) activations are included wherever possible, and dropout layers are added ($P = 0.6$) for imputation models. For each sample $\mathbf{x}_i$ and $\mathbf{y}_j$, we sample the multivariate Gaussian

$$[f_E^X(\mathbf{x})]_i = \begin{bmatrix} \boldsymbol{\mu}_i^X \\ \boldsymbol{\sigma}_i^X \end{bmatrix}, \quad L_i^X \sim \mathcal{N}(\boldsymbol{\mu}_i^X, \text{diag}(\boldsymbol{\sigma}_i^X)), \quad (2a)$$

$$\boldsymbol{\mu}_i^X, L_i^X \in \mathbb{R}^g, \quad \boldsymbol{\sigma}_i^X \in \mathbb{R}_{>0}^g, \quad i \in \{1, 2, \dots, n_x\},$$

$$[f_E^Y(\mathbf{y})]_j = \begin{bmatrix} \boldsymbol{\mu}_j^Y \\ \boldsymbol{\sigma}_j^Y \end{bmatrix}, \quad L_j^Y \sim \mathcal{N}(\boldsymbol{\mu}_j^Y, \text{diag}(\boldsymbol{\sigma}_j^Y)), \quad (2b)$$

$$\boldsymbol{\mu}_j^Y, L_j^Y \in \mathbb{R}^g, \quad \boldsymbol{\sigma}_j^Y \in \mathbb{R}_{>0}^g, \quad j \in \{1, 2, \dots, n_y\},$$

where $[f_E^X(\mathbf{x})]_i$ and $[f_E^Y(\mathbf{y})]_j$ are the encoded results of $\mathbf{x}_i$ and $\mathbf{y}_j$, respectively, and $\text{diag}(\cdot)$ is a diagonal matrix generated from a vector. L_i^X and L_j^Y are sampled from the multivariate Gaussian distributions indicated above. We define $L_i^X \in \mathbb{R}^g$ as the i th column of L^X , representing the feature vector of the i th sample. $L_j^Y \in \mathbb{R}^g$ is defined similarly.

The VAE is able to approximately map any non-Gaussian input data onto the multivariate Gaussian latent space (using the encoder) and reconstruct the input data from the space using the decoder²⁸. Therefore, the multimodal input to JAMIE is not restricted to a Gaussian distribution.

Aggregation of the latent spaces

Using our user-provided correspondence matrix $F \in \mathbb{R}^{n_x \times n_y}$, we can perform an aggregate calculation to combine the latent space vectors of known aligned points. In particular,

$$M^X = (L^X + \delta L^Y F^T) (\text{diag}(1 + \delta F \mathbf{1}_{n_y}))^{-1}, \quad (3a)$$

$$M^Y = (L^Y + \delta^{-1} L^X F) (\text{diag}(1 + \delta^{-1} F^T \mathbf{1}_{n_x}))^{-1}, \quad (3b)$$

where the aggregate feature vector $M^X \in \mathbb{R}^{g \times n_x}$ has g features and n_x samples and $M^Y \in \mathbb{R}^{g \times n_y}$ has g features and n_y samples. $\mathbf{1}_{n_x} \in \mathbb{R}^{n_x}$ and $\mathbf{1}_{n_y} \in \mathbb{R}^{n_y}$ are vectors of all ones. δ refers to the relative weighting between modalities X and Y in the aggregation function. We learn δ over the course of training. A value $\delta < 1$ implies that X is weighted more than Y during aggregation, while $\delta > 1$ implies the opposite. The construction of M^X, M^Y is shown visually in Fig. 2a in the formulation of the blue vector. Figure 2a shows the case $F_{ik} = 1$ with all other entries in row i and column k being 0, which results in simply averaging the latent feature vectors of $\mathbf{x}_i$ and $\mathbf{y}_k$.

The linear aggregation technique is backed by the non-linear transformation of the encoder, which is able to account for varying timings, magnitude changes and distributions owing to its deep structure. In Fig. 2, this is the blue latent space vector combination of correspondent samples. We call this technique correlation-based latent aggregation. This moves correspondent latent embeddings in similar directions over the course of training and is key in the formation of similar latent spaces. Correlation-based latent aggregation is the primary motivation for the use of the VAE framework, as it allows for this intuitive representation of partially aligned datasets. We can now adjust our low-dimensional data embeddings by using aggregates wherever possible to produce our final aggregate latent spaces M^X and M^Y .

Continuity of the aggregate latent spaces

Continuous L^X and L^Y facilitate imputation, thus we adopt the VAE architecture²⁷ and the corresponding KL loss:

$$\ell_{kl} = \frac{\kappa}{g} \sum_{k=1}^g d_k, \quad (4)$$

where d_k takes the KL divergence of the k th feature between the distribution of our latent space estimate and a multivariate standard normal as the targeted distribution:

$$d_k = -\frac{1}{2n_x} \sum_{i=1}^{n_x} \left(1 + 2 \log([\sigma_i^X]_k) - ([\mu_i^X]_k)^2 - ([\sigma_i^X]_k)^2 \right) - \frac{1}{2n_y} \sum_{j=1}^{n_y} \left(1 + 2 \log([\sigma_j^Y]_k) - ([\mu_j^Y]_k)^2 - ([\sigma_j^Y]_k)^2 \right). \quad (5)$$

The parameter κ is slowly increased from 0 to 1 over the course of training to accelerate convergence²⁹. This process is referred to below as KL annealing and is described in ‘Early Stopping and KL Annealing’.

ℓ_{kl} will lower to zero as the latent space’s distribution approaches the standard multivariate Gaussian. This encourages a continuous latent space through the non-zero s.d. The mean approaching zero encourages a small, clustered latent space. This will negatively affect metrics such as silhouette distance, but can aid in imputing outlier or unseen single-cell data.

Similarity of the original and aggregate latent spaces

The structure alone with JAMIE already produces shared latent spaces M^X and M^Y . To balance L^X and L^Y and keep both latent spaces the same scale, we implement a loss proportional to the square norm of the difference between each latent space and its aggregated matrix M :

$$\ell_{\text{combination}} = \frac{1}{gn_x} \|L^X - M^X\|_F^2 + \frac{1}{gn_y} \|L^Y - M^Y\|_F^2. \quad (6)$$

$\|\cdot\|_F$ indicates the Frobenius norm. The combination loss makes the aggregate matrices as similar as possible to the originals while still maintaining cross-modal corresponding cell representation similarity. In practice, as long as the two modalities contain unique information and are non-trivially corresponding, the combination and reconstruction losses will balance each other out, as the reconstruction loss ensures that information unique to each modality is encoded in the latent representations. This balance allows JAMIE to form the latent space flexibly while preserving as much single-modal detail as possible.

Similarity of the original and reconstructed data matrices

Decoders are structured in the reverse manner as encoders. It uses fully connected layers to transform the feature dimension as follows: $g \rightarrow n_x \rightarrow 2n_x \rightarrow n_x$. The structure is similar for Y . Batch normalization and leaky ReLU are included where possible, and dropout layers are added ($P = 0.6$) for imputation models. We denote the decoders $f_D^X(\cdot), f_D^Y(\cdot)$ for each modality and use each to get an estimate of the original datasets $\tilde{X} = f_D^X(M^X) \in \mathbb{R}^{d_X \times n_X}$ and $\tilde{Y} = f_D^Y(M^Y) \in \mathbb{R}^{d_Y \times n_Y}$. The quality of the reconstruction, and thereby the information retained in the latent space, can be evaluated by

$$\ell_{\text{reconstruction}} = \frac{1}{d_X n_X} \|X - \tilde{X}\|_F^2 + \frac{1}{d_Y n_Y} \|Y - \tilde{Y}\|_F^2. \quad (7)$$

Correspondence of the aggregate latent spaces

We find the optimal inferred correspondence matrix $\bar{F} \in \mathbb{R}_{\geq 0}^{n_X \times n_Y}$ by solving the following optimization problem:

$$\begin{aligned} \min_{\alpha, \bar{F}} & \|\alpha K_X - \bar{F} K_Y \bar{F}^T\|_F^2 \\ \text{s.t. } & \bar{F} \geq 0, \quad \alpha \geq 0, \\ & \bar{F} \mathbf{1}_{n_Y} = \mathbf{1}_{n_X}, \quad \bar{F}^T \mathbf{1}_{n_X} \leq \mathbf{1}_{n_Y}, \end{aligned} \quad (8)$$

where $[K_X, K_Y] = [\text{dist}(X), \text{dist}(Y)]$ are the intra-dataset distance. This generally uses geodesic distance and is similar to the quadratic assignment problem.

The procedure originates from an extension on Cui et al.³⁰, which optimized a similar expression but assumed one-to-one correspondence on inter-dataset cell pairs. Reference⁷ extended this to the concept of one-to-many correspondence between modalities and created the implementation in equation (8).

Once solved, F_{ij} denotes the probability that sample x_i is matched with sample y_j . The objective of $\bar{F}$ is to provide complete correspondence information. F will generally contain only one-to-one relationships, while true inter-modal relationships are often layered and complex. In practice, we may consider combining F and $\bar{F}$ by a weighted average if the user-provided correspondence is especially low (Algorithm 1).

To ensure that correspondent samples have similar aggregate latent representations, we introduce $\ell_{\text{alignment}}$. This metric relies on our inferred correspondence $\bar{F}$ rather than the user-provided correspondence matrix F . Unlike F , our inferred correspondence $\bar{F}$ contains information about samples that are likely to not correspond. We can utilize this dense information to shape our latent space such that contributing samples are clustered:

$$\ell_{\text{alignment}} = \frac{1}{gn_x} \|M^X - M^Y \bar{F}^T\|_F^2. \quad (9)$$

The alignment loss is primarily effective when the user-provided correspondence is weak (containing few highly correspondent pairs). This loss shapes the integrated latent spaces and aids in obtaining cross-modal latent similarity by inferring correspondence using unsupervised manifold alignment. Any unsupervised correspondence method may be substituted for $\bar{F}$.

Early stopping and KL annealing

Early stopping is performed by keeping track of the best ℓ_{total} over the course of training. JAMIE is given a minimum number of epochs (with a default of 2,500) over which time no early stopping can take place. Over this period, κ is gradually increased from 0 to 1. This concept originates from Bowman et al.²⁹. Once this period is over, the training continues until the algorithm does not produce a new best loss for a set number of epochs (with a default of 500) or until the maximum number of epochs is achieved (with a default of 10,000). This process is further detailed in Algorithm 1.

JAMIE algorithm

In Algorithm 1, datasets X and Y are provided pre-standardized along with an optional correspondence matrix F as described in the introduction of Methods. The user-given variables are $\tau_{\min}$ (with a default of 2,500) and $\tau_{\max}$ (with a default of 10,000), which are the minimum and maximum number of epochs for training. b_s (with a default of 512) is the batch size. α, β and γ are the weights for the model, with default values of 1,000, 31.25 and 31.25, respectively. $\tau_{\max_lapses}$ (with a default of 500) and $\ell_{\min_change}$ (with a default of 1×10^{-8}) control the behaviour of the early-stopping algorithm.

ρ is an experimental feature that controls how much the inferred correspondence matrix is utilized over the user-provided correspondence. The weighted average $\rho F + (1 - \rho) \bar{F}$ is used in equations (3a) and (3b). By default, $\rho = 1$. This is not explored in depth within the paper but is a useful feature for low-correspondence applications.

Performance evaluation

LTA. To assess phenotype (for example, cell type or cortical layer) separation, we use LTA^{7,19}. LTA measures the phenotype identification accuracy of a nearest-neighbour classifier trained on the second modality and evaluated on the first. Higher values generally indicate better phenotype separation.

Algorithm 1 JAMIE

Input: X, Y, F
Params: $\tau_{\min}, \tau_{\max}, b_s, (\alpha, \beta, \gamma), \rho, \tau_{\max_lapses}, \ell_{\min_change}$
Output: $f_E^X(\cdot), f_E^Y(\cdot), f_D^X(\cdot), f_D^Y(\cdot), L^X, L^Y$

- 1: $[K_X, K_Y] = [\text{dist}(X), \text{dist}(Y)]$
- 2: $\tilde{F} = \arg \min_{\alpha, \tilde{F}} \|\alpha K_X - \tilde{F} K_Y \tilde{F}^T\|_F^2$
- 3: $F \leftarrow \rho F + (1 - \rho) \tilde{F}$
- 4: $\ell_{\text{best}} \leftarrow \infty, \tau_{\text{lapses}} \leftarrow 0$
- 5: $\tau_{\text{current}} \leftarrow 0$
- 6: Initialize $f_E^X, f_E^Y, f_D^X, f_D^Y$ with random weights
- 7: **while** $\tau_{\text{current}} < \tau_{\max}$ and $\tau_{\text{lapses}} < \tau_{\max_lapses}$ **do**
- 8: Randomly sample $X_s \in \mathbb{R}^{d_X \times b_s}$ from $X \in \mathbb{R}^{d_X \times n_X}$ and $Y_s \in \mathbb{R}^{d_Y \times b_s}$ from $Y \in \mathbb{R}^{d_Y \times n_Y}$
- 9: Extract submatrix $F_s \in \mathbb{R}^{b_s \times b_s}$ from $F \in \mathbb{R}^{n_X \times n_Y}$ and $\tilde{F}_s \in \mathbb{R}^{b_s \times b_s}$ from $\tilde{F} \in \mathbb{R}^{n_X \times n_Y}$
- 10: Optionally transform X_s, Y_s using PCA trained on X, Y
- 11: Normalize columns of F_s and $\tilde{F}_s$ to summation 1
- 12: Randomly sample $L_i^{X_s}$ from distributions $f_E^X(X_s)$ and $L_j^{Y_s}$ from distributions $f_E^Y(Y_s)$ for $i, j \in \{1, 2, \dots, b_s\}$
- 13: $M^{X_s} \leftarrow (L^{X_s} + \delta L^{Y_s} F_s^T) (\text{diag}(1 + \delta F_s \mathbf{1}_{b_s}))^{-1}$
- 14: $M^{Y_s} \leftarrow (L^{Y_s} + \delta^{-1} L^{X_s} F_s) (\text{diag}(1 + \delta^{-1} F_s^T \mathbf{1}_{b_s}))^{-1}$
- 15: $\tilde{X}_s \leftarrow f_D^X(M^{X_s})$ and $\tilde{Y}_s \leftarrow f_D^Y(M^{Y_s})$
- 16: If X_s, Y_s were transformed using PCA, perform the inverse transform on $\tilde{X}_s, \tilde{Y}_s$
- 17: $\kappa \leftarrow \left(1 + \exp\left(\frac{-5(\tau_{\text{current}} - 0.5\tau_{\min})}{0.5\tau_{\min}}\right)\right)^{-1}$
- 18: Calculate ℓ_{kl} via equation (4)
- 19: Calculate $\ell_{\text{combination}}$ via equation (6)
- 20: Calculate $\ell_{\text{reconstruction}}$ via equation (7)
- 21: Calculate $\ell_{\text{alignment}}$ via equation (9)
- 22: $\ell_{\text{total}} \leftarrow \ell_{\text{kl}} + \alpha \ell_{\text{combination}} + \beta \ell_{\text{reconstruction}} + \gamma \ell_{\text{alignment}}$
- 23: **if** $\tau_{\text{current}} \geq \tau_{\min}$ **then**
- 24: **if** $\ell_{\text{total}} < \ell_{\text{best}} - \ell_{\min_change}$ **then**
- 25: $\ell_{\text{best}} \leftarrow \ell_{\text{total}}$
- 26: $\tau_{\text{lapses}} \leftarrow 0$
- 27: **else**
- 28: $\tau_{\text{lapses}} \leftarrow \tau_{\text{lapses}} + 1$
- 29: **end if**
- 30: **end if**
- 31: Update $f_E^X, f_E^Y, f_D^X, f_D^Y$ using gradient descent on ℓ_{total}
- 32: $\tau_{\text{current}} \leftarrow \tau_{\text{current}} + 1$
- 33: **end While**
- 34: $L^X \leftarrow f_E^X(X), L^Y \leftarrow f_E^Y(Y)$

FOSCTTM. We use this metric to evaluate the average fraction of samples closer to the truly aligned sample (in both modalities) than its actual pairing. This provides an estimate of how closely the two modalities are aligned. Lower values generally indicate closer cross-modal alignment.

Pearson correlation. We use Pearson correlation mainly to compare imputed and measured feature values, providing an estimate of how well the imputation method captures the variance of the measured feature values. For the purposes of imputation, higher values usually indicate better performance.

AUROC. We use the AUROC to evaluate imputed features for prediction of high or low measured expression values. To test this, we use AUROC

on the median-binarized measured data. That is, each feature value above the feature median is treated as 1, and all others are treated as 0.

JS distance. It is also important that an imputation method preserve individual cell-type distributions. We use the JS distance, which is the square root of the JS divergence. The JS divergence is the average KL divergence between two distributions and their average (a symmetrized version of the KL divergence). The KL divergence measures the difference between two probability distributions, and increases proportionally. As a baseline for imputation performance, a KNN model ($k = 5$) is used to generate new multimodal data based on the average of nearest neighbours in the training set.

P values. When we compare two methods in terms of imputation performance, we utilize the two-tailed binomial test with the null hypothesis that both methods have an equal probability of imputing better on any given feature.

Hyperparameters and validation

PCA is used to lower the dimensionality of data with many features. The default value for this preprocessing is 512 features, and is automatically inverted when applying imputation. The hyperparameters used for all experiments in the paper are the default losses detailed in the introduction of Methods, the minimal and maximal iterations detailed in ‘Early stopping and KL annealing’ and a batch size of 512.

When performing integration in Results, no data are withheld, hyperparameters are left at default and dropout is set to $P = 0$ to facilitate maximal learning of single-modal nuances, regardless of dataset specificity. This is done because many alignment methods are unable to be reused, and perform integration only once before having to be recalculated. For imputation, hyperparameters are left unchanged but dropout is set to $P = 0.6$ to facilitate generalization. Of cells within each dataset, 20% are withheld at random. Any imputation performance calculations, such as those in Fig. 3f, are performed on these withheld data.

Datasets and preprocessing

We applied JAMIE to four main single-cell multimodal datasets: (1) simulated multimodal data generated by sampling from a Gaussian distribution on a branching manifold in MMD-MA ($N = 300$, three cell types)⁸, (2) Patch-seq gene expression and ephys features of single neuronal cells in the mouse visual cortex ($N = 3,654$, six cell types)² and mouse motor cortex ($N = 1,208$, nine cell types)³¹, (3) 10x Multiome gene expression and chromatin accessibility data of 8,981 cells in the developing human brain (21 postconceptional weeks, covering seven major cell types in human cerebral cortex)³ and (4) scRNA-seq gene expression and scATAC-seq chromatin accessibility data of 4,301 cells from the COLO-320DM colon adenocarcinoma cell line (which was also analysed by BABEL⁵)²². The input datasets to JAMIE were standardized (mean 0, s.d. 1) across features, and the dimensionality of the large numbers of input features was reduced by PCA to 512 PCs as needed for efficient computation.

Applications

Phenotype prediction from multimodal data. Integration on multimodal datasets can improve classification, knowledge of phenotype and understanding of complex biological mechanisms. Given two datasets X, Y and correspondence F , we generate L^X, L^Y from the trained encoders $f_E^X(\cdot), f_E^Y(\cdot)$ as in ‘Distribution of latent space features’. Clustering can be performed, or classification methods may be directly built atop the reusable latent space. Clustering on these latent spaces confers several advantages, the main one being the incorporation of both modalities in the process of feature generation. We can then actively predict which samples correspond (if not already known) and perform related tasks such as cell-type prediction. On partially annotated datasets, cells in similar clusters should be of similar cell type.

There is generally no need to use a complicated clustering or classification algorithm, as JAMIE should do most of the separation as a part of the latent space generation. The networks may be further analysed to provide a clearer picture of the relationship between data features and phenotype. To visualize this cell-type clustering, UMAP³² is performed on high-dimensional data for visualization.

Cross-modal imputation. There are several methods to perform cross-modal imputation, but many methods are not able to prove that they have learned underlying biological mechanisms for the purpose of imputation. When utilized for cross-modal imputation, we can predict our missing data with a more rigorous foundation than that offered by a feed-forward network or linear regression. Given training data X, Y , we can train the model. With new data $X^* \in \mathbb{R}^{d_x \times n_{x^*}}$, we can predict its correspondent latent embedding $\hat{Y} = f_D^Y(L^{X^*}) \in \mathbb{R}^{d_y \times n_{x^*}}$, which has true values Y . Then, we have predicted correspondent cells using a latent space that probably confers understanding of the relationship between data features and phenotype.

This has motivation within the given losses. The alignment loss incentivizes the low-dimensional embeddings for each modality to be on the same latent space. Additionally, since we are predicting correspondent cells, $F = \mathbb{I}$. Ideally then, $L^{X^*} = M^{X^*} = M^Y = L^Y$. So, if we assume perfect and total alignment, $f_D^Y(L^{X^*}) = f_D^Y(L^Y)$. This prediction is shown in Fig. 2b. Of course, this ideal scenario is not always the case. The efficacy of this cross-modal imputation method is explored within Results, where 20% of the data is taken for validation and the model is trained on the remainder.

Latent and imputed feature interpretation. For interpretation of our trained model, we employ the method described in Lundberg et al.¹⁸, referred to as SHAP. SHAP looks at individual predictions produced by a model and will assess the importance of various input features through sample modulation. This can be used for a variety of interesting applications. If a target variable is easily separable by phenotype, SHAP can determine relevant features for further study. Additionally, if we perform imputation, SHAP can expose inter-modal linkages learned by the model. Given a model f and a sample $x \in \mathbb{R}^n$, SHAP values $\phi_0, \dots, \phi_{n-1}$ are learned such that $\mathbb{E}[f(z)] + \sum_{i \in \{j, \dots, m\}} \phi_i = \mathbb{E}[f(z)|z_j = x_j, \dots, z_m = x_m]$ for a background feature vector $z \in \mathbb{R}^n$. If $\{j, \dots, m\} = \{0, 1, \dots, n-1\}$, then the sum of SHAP values and the background output will equal $f(x)$, with each ϕ_i proportional to the impact of x_i on the model output.

Another useful technique involves choosing a key metric for classification (for example, LTA) or imputation (for example, the correspondence between imputed and measured features) and evaluating the metric with each feature sequentially removed (replaced with its background) from the model. Then, if the key metric becomes worse, it suggests that the removed feature is more important to the outcome of the model.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The MMD-MA simulation dataset can be downloaded from <https://noble.gs.washington.edu/proj/mmd-ma/>. Our simulation data may be downloaded from <https://github.com/daifengwanglab/JAMIE>. Processed Patch-seq gene expression and electrophysiological features for the mouse visual and motor cortices are available at <https://github.com/daifengwanglab/scMNC>. Raw Patch-seq datasets are available at refs. 2,31. Single-cell RNA-seq and ATAC-seq data on the developing human brain can be downloaded at <https://github.com/GreenleafLab/brainchromatin/blob/main/links.txt> under the heading ‘Multiome’. Single-cell RNA-seq and ATAC-seq of colon adenocarcinoma data can

be found at <https://github.com/wukevin/babel>. Processed datasets for SNARE-seq adult mouse cortex data can be downloaded from <https://scglue.readthedocs.io/en/latest/data.html>.

Code availability

All code was implemented in Python using PyTorch, and the source code is publicly available at <https://github.com/daifengwanglab/JAMIE>³³. Since Code Ocean provides an interactive platform for computational reproducibility³⁴, we have also provided an interactive version of our code for reproducing results and figures³⁵.

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Author contributions

D.W. conceived and supervised the study. N.C.K. developed and implemented the methodology. X.H. and D.W. verified the methods. N.C.K. performed visualization and analysis. N.C.K., X.H. and D.W. edited and wrote the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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